

SAI LELLA

Senior Data Engineer

Sailella800@gmail.com | 940-445-6361 | linkedin.com/in/sai-lella-99a7b9169/

PROFESSIONAL SUMMARY

- **Senior Data Engineer** with **5+ years** architecting scalable, **HIPAA-compliant cloud data platforms** across **healthcare and banking** domains, delivering enterprise-grade **Lakehouse architectures** using **Azure Databricks** and **Microsoft Fabric** to support regulatory reporting and production-grade data foundations for predictive analytics.
- Engineered **high-velocity batch and real-time pipelines** processing **1B+ records monthly**, implementing **CDC-based ingestion** and **Spark optimization** that reduced pipeline runtimes by **30%**.
- Architected **feature engineering pipelines** and **governed data layers**, transforming transactional datasets into **model-ready inputs** for **risk scoring**, **predictive analytics**, and **Azure Machine Learning** workflows.
- Streamlined production deployments using **CI/CD (Azure DevOps, Terraform)** and containerization (**Docker, Kubernetes**), reducing development effort by **25%** and improving SLA adherence across enterprise data platforms.
- Implemented **automated data quality frameworks**, **schema enforcement**, and **governance controls**, enabling audit-ready data pipelines aligned with **HIPAA** and **banking compliance** standards.
- Defined reusable **Spark framework standards** and **Lakehouse design patterns**, improving cross-team development efficiency and ensuring scalable, maintainable enterprise data architecture.
- Partnered with **analytics, compliance, and data science teams** to deliver secure, high-performance data solutions powering **executive dashboards**, **regulatory reporting**, and **AI-driven decision systems**.

TECHNICAL SKILLS

Cloud Data Platforms: Azure (Databricks, Microsoft Fabric, Synapse, Data Lake Gen2), AWS (S3, EMR, Redshift, Glue, Athena, Lambda), Snowflake

Data Engineering & Big Data: Apache Spark, PySpark, Spark SQL, Delta Lake, Apache Iceberg, Medallion/Lakehouse Architecture, Hive, HDFS, YARN, CDC Frameworks

AI & ML Data Enablement: Feature Engineering Pipelines, Model-Ready Dataset Preparation, MLflow (Experiment Tracking), Azure Machine Learning, Data Foundations for LLM & Predictive Analytics

Data Governance & Observability: Data Lineage, Metadata Management, SLA Monitoring, Schema Enforcement, Role-Based Access Control (RBAC), HIPAA & Banking Compliance

Integration & Orchestration: Azure Data Factory, Microsoft Fabric Data Factory, Apache Airflow, AWS Glue, dbt, Fivetran, Matillion

Streaming & Real-Time: Apache Kafka, Spark Structured Streaming, Event-Driven Architectures

Languages & Databases: Python, SQL, PySpark, Scala, Java; SQL Server, PostgreSQL, Oracle, Teradata, DB2, MongoDB, Cassandra, HBase

DevOps & Platform Engineering: Terraform (IaC), Docker, Kubernetes, Jenkins, Azure DevOps, Git-based CI/CD, AWS CloudFormation

Analytics & BI: Power BI (DAX, Semantic Models), Tableau, SQL-based Reporting

PROFESSIONAL EXPERIENCE

U.S. Bank | Dallas, TX

Dec 2024 – Present

Senior Data Engineer

- Designed and maintained **enterprise-scale ELT pipelines** using **Microsoft Fabric Data Factory** and **Azure Databricks**, processing **1B+ financial transaction records monthly** into governed Lakehouse environments.
- Engineered scalable **Spark/PySpark pipelines** handling **multi-terabyte** structured and semi-structured banking datasets supporting **transaction analytics**, **loan reporting**, and risk analysis.
- Implemented **CDC-based ingestion frameworks** across core banking systems, reducing full-load dependencies by **40%** and improving data freshness and SLA adherence.

- Developed **SCD Type 1 & Type 2 transformations** and dimensional models supporting **SOX-aligned financial reporting** and audit traceability.
- Architected and managed **Medallion (Bronze/Silver/Gold) Lakehouse architecture**, enabling curated datasets for executive dashboards, compliance reporting, and advanced analytics.
- Built scalable **feature engineering pipelines** transforming transactional and loan data into **model-ready inputs** supporting **credit risk scoring** and predictive analytics initiatives.
- Integrated curated data layers with **Azure Machine Learning environments** to enable batch training datasets and production inference data pipelines for risk analytics workflows.
- Supported AI-driven initiatives by preparing structured and governed datasets for **LLM-assisted analytics** and decision-support use cases, ensuring secure and compliant data access.
- Implemented enterprise-grade data governance controls including **RBAC, schema enforcement, data quality monitoring**, and audit logging, aligned with banking regulatory standards.
- Optimized Spark workloads through partitioning strategies, caching, and cluster autoscaling, reducing runtime by **30%** across high-volume financial datasets.
- Established **data lineage tracking, SLA monitoring**, and observability frameworks, improving pipeline transparency and operational reliability across business-critical data assets.
- Standardized reusable **Spark frameworks and Lakehouse design patterns**, reducing development effort by **25%** and improving engineering scalability.

Environment: Microsoft Fabric (Data Factory, Lakehouse, OneLake), Azure Databricks, Azure Data Lake Gen2, Azure Synapse Analytics, Azure Machine Learning, Snowflake, Python, PySpark, Spark SQL, Delta Lake, SQL Server, CDC Frameworks, Hive, Tableau, Power BI

Cigna Healthcare | Plano, TX

Dec 2023 – Nov 2024

Data Engineer

- Designed and developed scalable **Azure-based data warehousing solutions** using **Azure Databricks, Azure Data Factory**, and **Azure Synapse Analytics**, implementing end-to-end **ETL/ELT pipelines** for healthcare analytics.
- Built **Spark/PySpark pipelines** processing **500M+ healthcare claims** and member enrollment records monthly, delivering curated datasets to enterprise reporting and analytics teams.
- Led migration of legacy healthcare data systems to **Azure cloud**, ensuring secure **PHI handling, role-based access control (RBAC)**, and **HIPAA-compliant** data processing.
- Developed transformation workflows and **dimensional data models** supporting **claims analytics, risk adjustment**, and **HEDIS reporting** initiatives.
- Implemented **CDC-based ingestion frameworks** to improve data freshness and reduce full-load dependencies across claims and provider data systems.
- Architected **Medallion (Bronze/Silver/Gold) Lakehouse layers** to enable scalable **population health analytics** and regulatory reporting.
- Built **feature engineering pipelines** transforming claims and clinical datasets into **model-ready inputs** for predictive analytics and **risk stratification** use cases.
- Integrated curated datasets with **Azure Machine Learning environments** to support batch training and inference workflows for **population health and cost prediction models**.
- Implemented enterprise-grade **data quality frameworks, schema validation**, and monitoring mechanisms, ensuring reliable and audit-ready healthcare datasets.
- Optimized **Spark workloads and Delta Lake storage formats**, reducing pipeline runtimes by **25%** and improving cost efficiency across large-scale healthcare data processing.
- Collaborated with data analysts, BI teams, and healthcare stakeholders to deliver secure, governed datasets powering **executive dashboards** and **population health insights**.
- Established **SLA monitoring** and pipeline observability practices, improving operational transparency and reducing production data failures.

Environment: Azure Databricks, Azure Data Factory, Azure Synapse Analytics, Azure Data Lake Gen2, Azure Machine Learning, Snowflake, dbt, Apache Kafka, Spark, PySpark, Delta Lake, SQL Server, CDC Frameworks, Hive, Power BI, Tableau

Cognizant | Hyderabad, India

Aug 2020 – Aug 2023

Data Engineer

- Architected and supported scalable **AWS-based data platforms** using **S3, EMR, Redshift, Glue, and Lambda**, enabling enterprise-grade data processing across multiple business domains.
- Designed and optimized **Redshift data warehouses** supporting **20+ analytical datasets**, processing **800GB+ daily** for reporting and operational analytics.
- Built batch and streaming data pipelines using **Apache Spark, PySpark, and Kafka**, integrating structured and semi-structured datasets into **Snowflake and Redshift** environments.
- Implemented **AWS Glue ETL workflows** and Python-based transformations, improving pipeline reliability and reducing data processing latency by **20%**.
- Orchestrated end-to-end workflows using **Apache Airflow**, managing dependencies across **30+ production pipelines** with SLA-based monitoring.
- Led data migration initiatives from **Teradata, DB2, and Oracle** into **Amazon S3 and Snowflake**, modernizing legacy systems and improving query performance by **35%**.
- Implemented **Infrastructure as Code (Terraform & AWS CloudFormation)** to provision scalable EMR clusters and data infrastructure, improving deployment consistency and reducing environment setup time by **40%**.
- Developed **data quality validation checks** and failure-handling mechanisms across batch and streaming pipelines, ensuring production-grade reliability and audit readiness.
- Optimized **Spark and YARN configurations** through memory tuning and partition strategies, reducing job runtimes by **25%** across high-volume datasets.
- Built ingestion pipelines using **Fivetran and Matillion**, streamlining integration of ERP and operational systems into **Snowflake data warehouses**.
- Collaborated with cross-functional teams including analytics, QA, and business stakeholders to deliver curated datasets supporting **executive reporting** and business intelligence initiatives.

Environment: AWS, Sqoop, SQL, Teradata, EMR, Redshift, Snowflake, Hive, Kafka, Glue, Apache Airflow, Docker, Kubernetes, Terraform, Jenkins, Tableau, Python, PySpark, AWS CloudFormation

EDUCATION

Master's in information technology

May 2025

Indiana Wesleyan University